Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
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1 Introduction

MERLS is a telehealth platform for assessing the speech of bilingual children at home. Patients complete speech assessments online, and speech clinicians use the platform to review the recordings, verify the results, and manage their patients' records and analytics. MERLS streamlines the clinician's workflow by combining speech annotation and assessment analysis in one system. Annotation is supported by automatically generated transcripts that clinicians can review and correct, and analysis is supported by organized, visualized results that point out behaviors needing the clinician's attention.

Because MERLS handles sensitive health information and supports clinical decisions about children, failures in the system could have serious consequences. A hazard is defined as any condition or state of the system that, under certain circumstances, could lead to harm to patients, such as an incorrect clinical conclusion or the disclosure of sensitive information. This document identifies the hazards in MERLS, analyzes how they could arise, and describes how the system's design and requirements reduce them.

2 Scope and Purpose of Hazard Analysis

This document examines the hazards in each component of MERLS outlined below. For each hazard, the analysis determines the possible failures, along with their effects, causes, and recommended actions. It covers both technical hazards and user interaction hazards, because clinicians rely on the system's outputs when assessing a child's language ability, and the information it stores is confidential. If these hazards are not addressed, patients and clinicians could suffer losses such as the exposure of private health information, incorrect clinical conclusions that affect a child's care, the permanent loss of assessment data, and a loss of confidence in the platform.

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

- Claude Sonnet 5 (Medium)
- ChatGPT-5.6 Luna

8.2 How and Where Used

Both models were used in all the sections to do the following:

- Check that each written section answered every question asked and met the rubric requirements
- Improve the wording, clarity and the flow of written sections
- Debug LaTeX compilation errors

The models were used to do the following in specific sections

- Brainstorm possible failure modes for the system components in section 5

8.3 Verification of AI-Generated Content

To verify the output of the LLM, each team member went over their section and presented their work to the group, explaining what they had completed since the previous meeting. Each member was able to successfully explain their sections. Additionally, before any pull request was approved, a team member who had not written the section reviewed the work to ensure it was cohesive, accurate, and consistent with the rest of the document.

The factual content of the documents came from the research of the group members. The writer of each section researched, obtained, and documented all referenced sources. After the models suggested wording improvements, the writer re-read the section to confirm that it still conveyed the same point and that the research was not altered.

9 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Question 1

Nour

What went well with this deliverable was determining possible failure modes regarding the ML Transcription model. There are several things to be wary of when dealing with external APIs and using machine learning models that have been tuned for our case. This is because they are not always predictable. Models typically have some form of evaluation, and we were given the model already chosen by our supervisors and seeing that it has not been evaluated was a bit concerning since it is hard to be certain how well such a model will perform.

What reassures me is that speech clinicians can go back and verify the model's output against the video playback, which is extremely important. However, this places the responsibility on the clinician, including how much time they are willing to spend annotating and whether they will watch the entire assessment instead of skimming through it.

Another worrying issue is data retention. If Zero Data Retention was not applied for when the team started using OpenAI's Whisper model, client information is being stored by OpenAI. OpenAI doesn't use this data to train its models unless a user explicitly agrees to share it, but the data is still stored on its servers. OpenAI is a large and popular company that could experience cyber attacks, so having sensitive data stored in multiple places is not ideal.

Question 2

This deliverable was relatively straightforward, and our team did not run into any major issues while completing it. We began by brainstorming the components of the system together, and we were able to identify what could go wrong with each one and possible hazards that could result. MERLS is a privacy-critical system, so security and data breaches were extremely important for us to address. The system also depends on performance and quality for different features to work, which made it easy to determine its high-level components. Because we share a strong understanding of the system, prioritizing the safety requirements was not difficult. Our supervisors also mentioned which parts of the system they wanted completed first, which gave us a clear direction as to what were the most important parts of the system.